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Foundation of Mathematics

TMS 1113

NUMBERS

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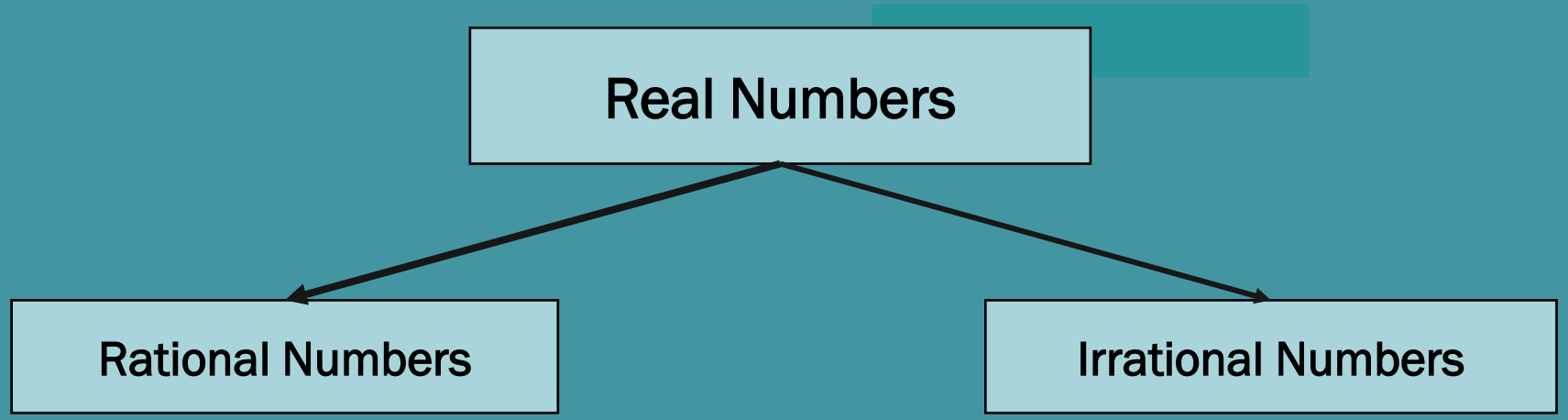
Objectives

- To identify the parts of the Real Number System.
- To identify the properties of numbers.
- To define rational and irrational numbers.
- To identify the basic arithmetic, fractions, decimals, percentages, etc.
- To classify numbers as rational or irrational

Numbers

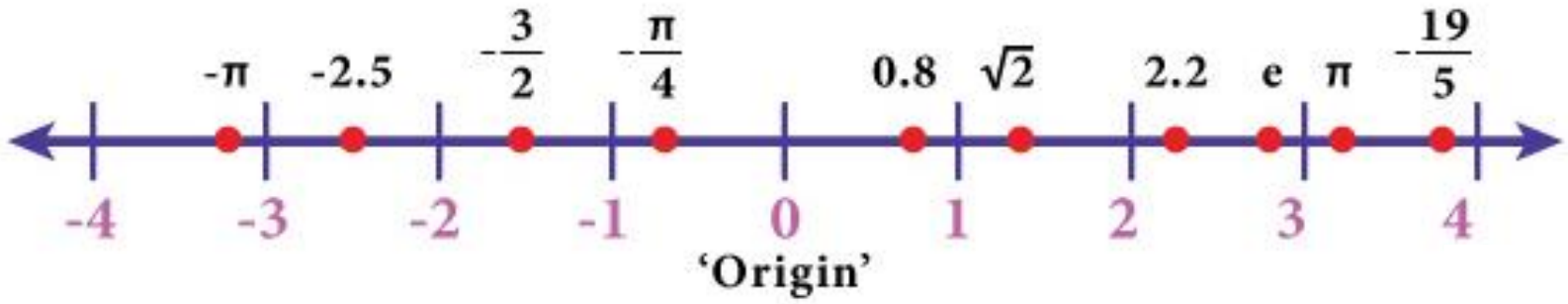
➢ What is a Number....?

- The number is an arithmetic value used to represent quantity.
- A number is a mathematical concept used to count, measure, and label.



Real Numbers

- Each real number corresponds to exactly one point on the number line, and every point on the number line represents only one real number.

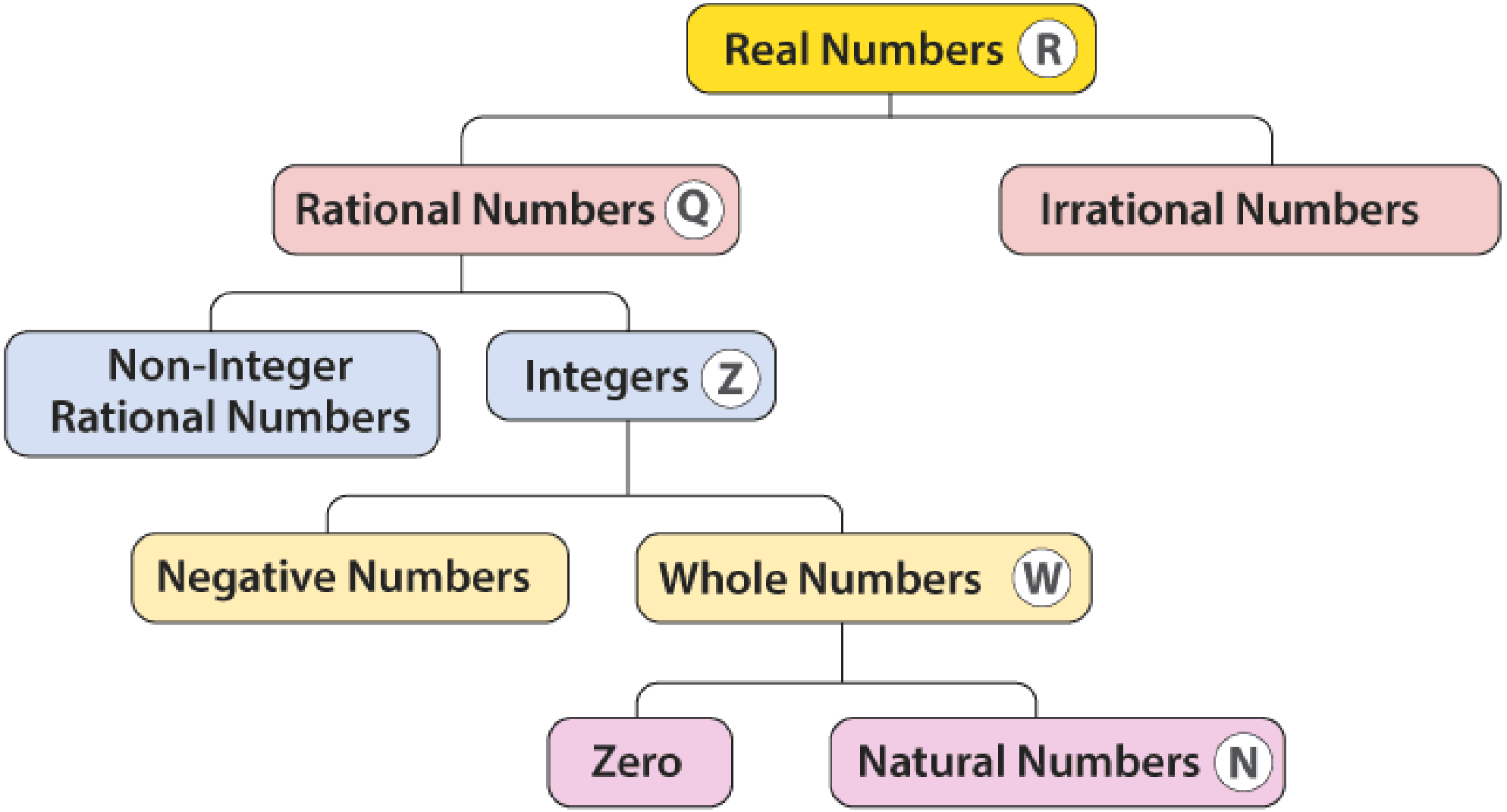


- All of the numbers that you use in everyday life are real numbers.

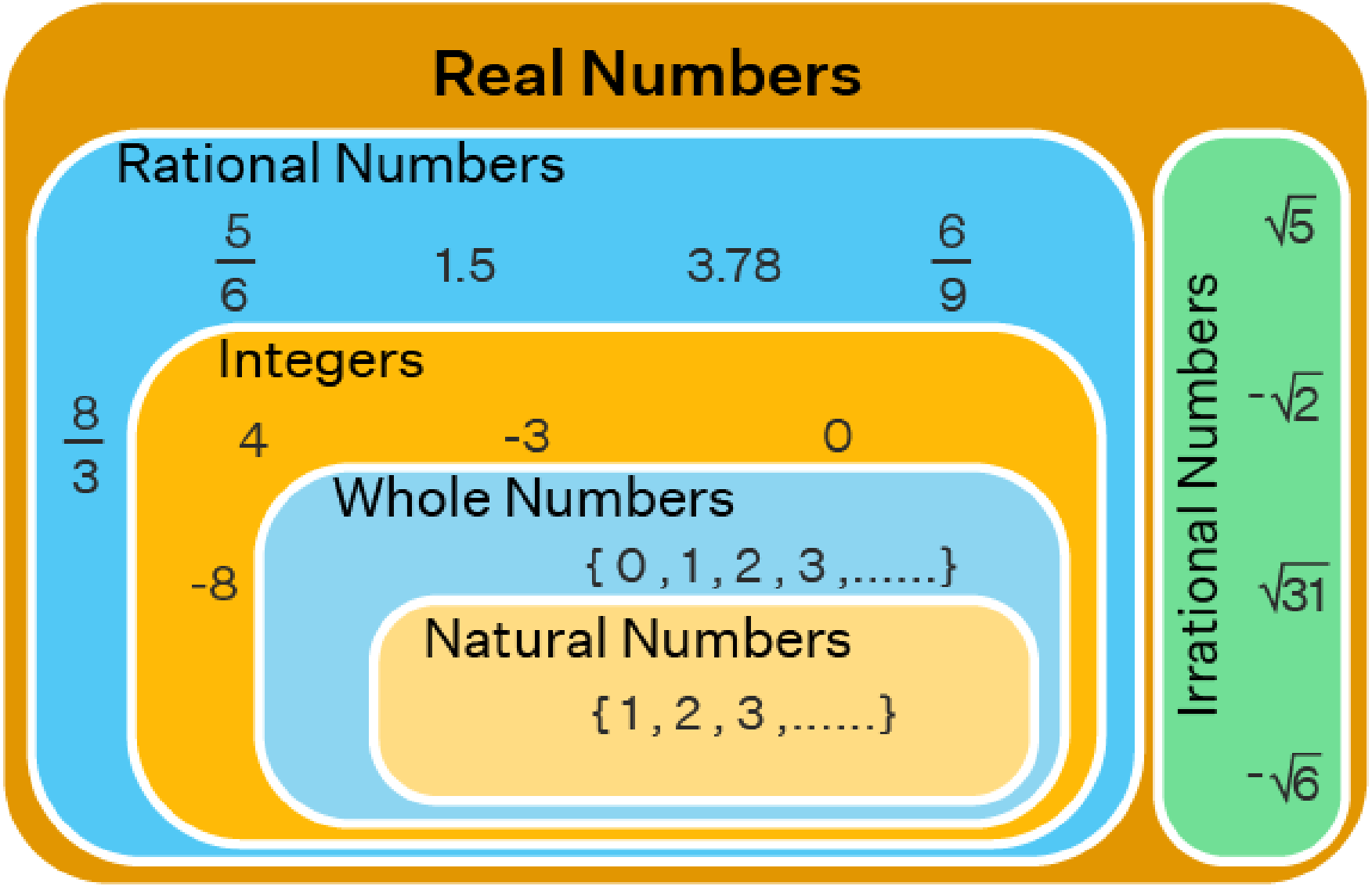
Example

154,769,852,35 4	-8	5,632.1010101256849765	49 %	π	$\sqrt{45}$	$\sqrt[5]{225}$

Real Number System Tree Diagram



What Comprises of Real Numbers?



Rational Number

- Real numbers can be classified either rational irrational.
- Rational numbers can be expressed as a ratio , where “a” and “b” are integers and b is not zero.
- The decimal form of a rational number is either a terminating or repeating decimal.

Example

Ratio Form	Decimal Form
$\sqrt{9}$	3.0
$\frac{3}{8}$	0.375
$\frac{3}{7}$	0.42857142857142857 or $\overline{0.428571}$

Irrational Number

- A real number that is not rational is an irrational number.
- An irrational number is a number that cannot be written as a fraction of two integers
- The decimal form of an irrational number neither terminates nor repeats

Example

$\Pi \approx 3.141592654\ldots\ldots$

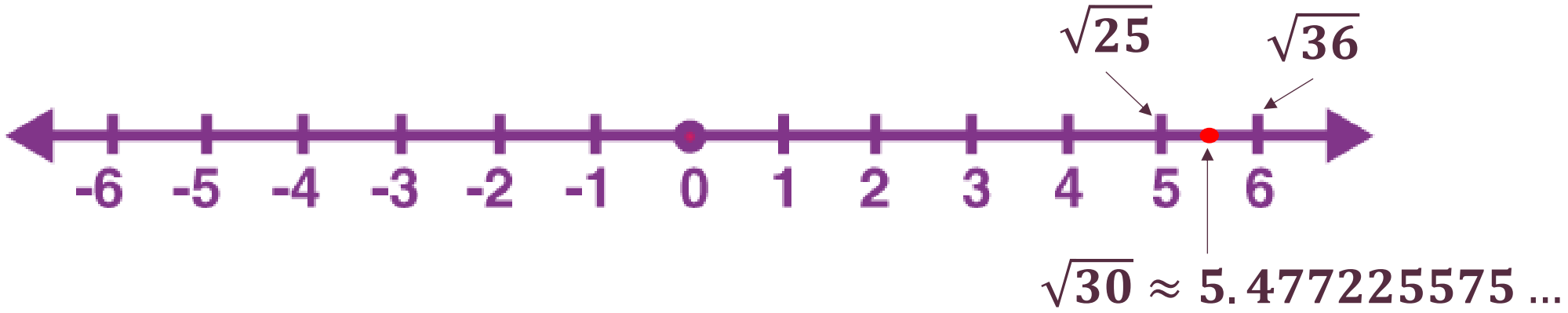
$e \approx 2.718281828\ldots\ldots$

$\sqrt{2}\ \ \sqrt{3}\ \ \sqrt{5}\ \ \sqrt{7}\ \ \sqrt{11}\ \ \sqrt{13}$

Do you notice a pattern within this group of numbers?
They're all PRIME numbers.....! A prime number only has two factors: 1 and itself

Irrational Number

- The square root of a positive integer is either a positive integer or an irrational number.
- For example, $\sqrt{36}$ is a positive integer, but $\sqrt{30}$, since it lies between 5 and 6, must be irrational.



Common Misconception

Do not assume that a number is irrational just because it is expressed using the square root symbol. Find its value first!

Example 1

1) $\sqrt{2}$

2) $\sqrt{12}$

3) $\sqrt{25}$

4) $\frac{5}{11}$

5) $\sqrt{66}$

a) Irrational
b) Irrational
c) Rational
d) Rational
e) Irrational

Study Tip

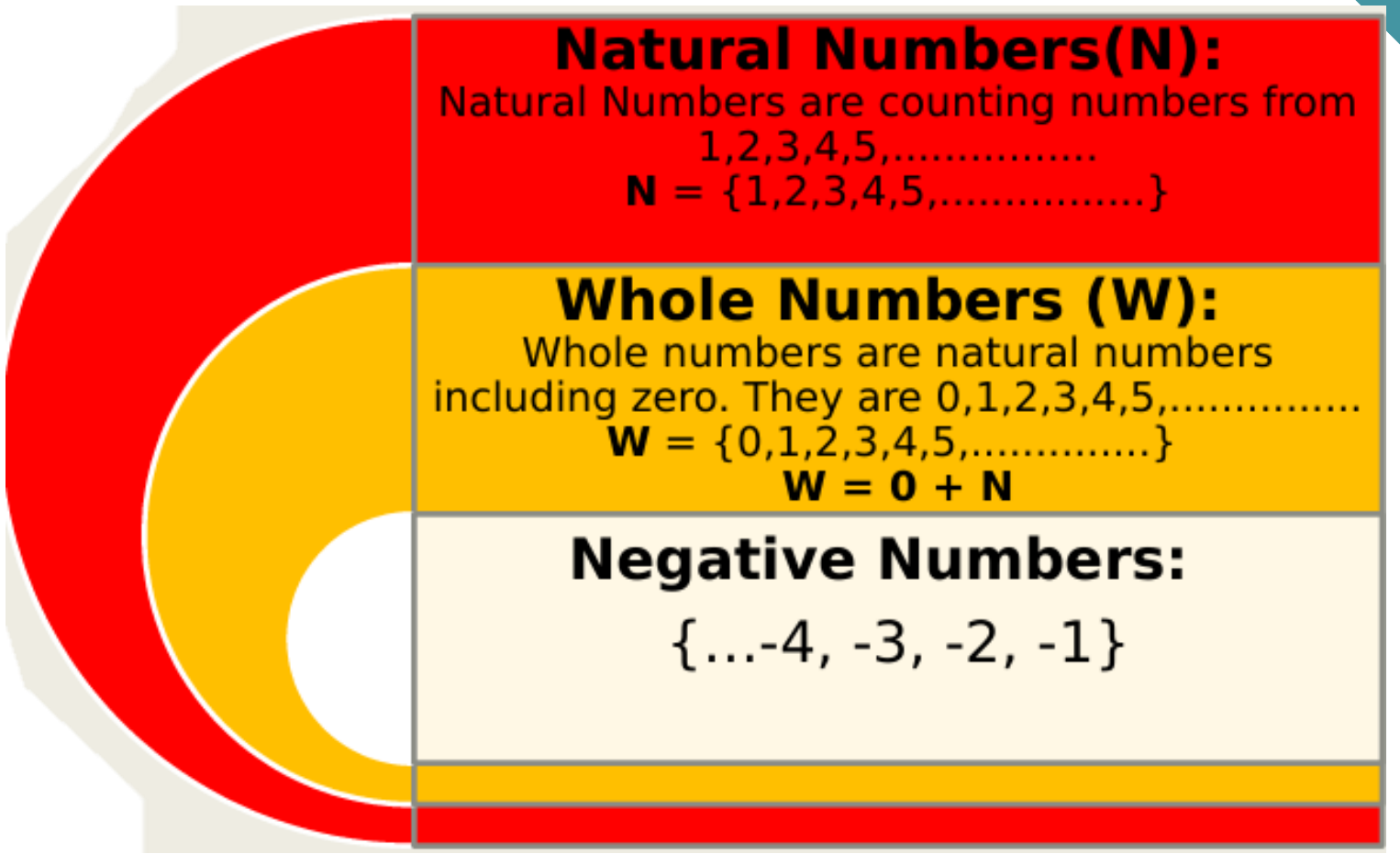
Recognize (at least) these numbers.

$\sqrt{4}$ $\sqrt{9}$ $\sqrt{25}$ $\sqrt{36}$ $\sqrt{49}$ $\sqrt{64}$ $\sqrt{81}$ $\sqrt{100}$ $\sqrt{121}$ $\sqrt{144}$ $\sqrt{169}$

Real Number System Tree Diagram

- Integers are natural numbers with zero and their negative values.
- They have a neutral number zero, the positive numbers with their negative counterparts.

Types of Integers



Example 2

Write all classifications that apply to each number…?

1) $\sqrt{5}$ Irrational, real- 5 is a whole number that is not a perfect square.

2) -12. 75 Rational, real–12.75 is a terminating decimal.

3) $\frac{\sqrt{16}}{2}$ Whole, integer, rational, real - $\frac{\sqrt{16}}{2} = \frac{4}{2} = 2$

Example 3

State whether the given numbers are rational, irrational, or not a real number…?

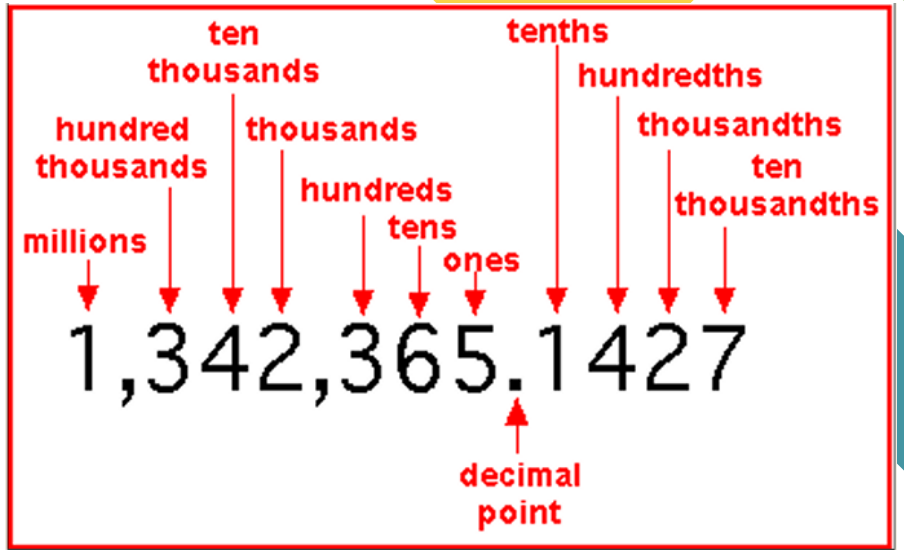
1) $\sqrt{21}$ Irrational

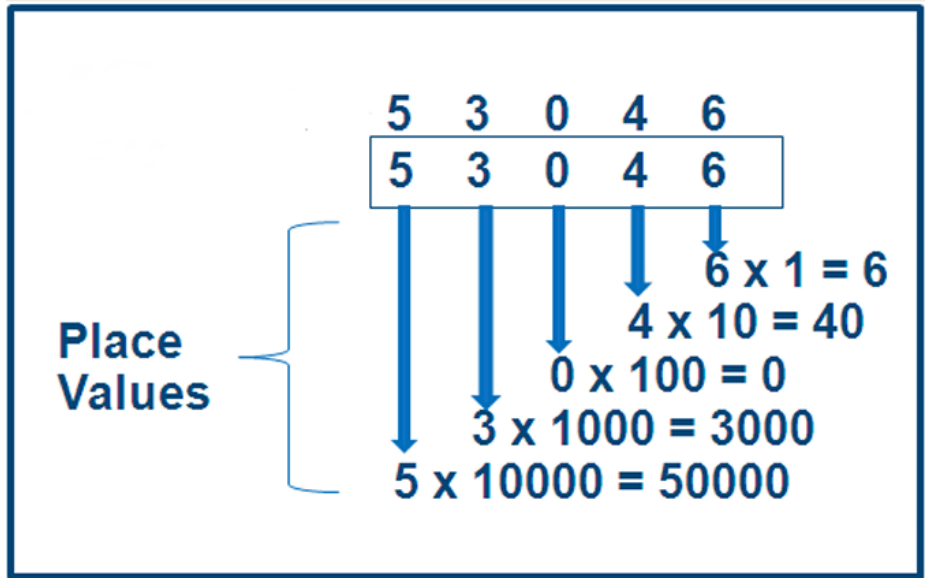
2) $\frac{0}{3}$ Rational $-\frac{0}{3} = 0$

3) $\frac{4}{0}$ Not a real number

Place Value of a Number

Whole Number Part						Fractional Part			
Ten Thousands	Thousands	Hundreds	Tens	Ones	← Decimal Point	tenths	hundredths	thousandths	ten thousandths
TTh	Th	H	T	O	.	t	h	th	tth
10,000	1,000	100	10	1	.	$\frac{1}{10}$	$\frac{1}{100}$	$\frac{1}{1,000}$	$\frac{1}{10,000}$





Even & Odd Numbers

- Even numbers are numbers that can be divided **evenly by 2.** Even numbers can be shown as a set like this:
  - {.....-4, -2, 0, 2, 4, 6,.....}
- Odd numbers are numbers that **cannot be divided evenly by 2.** Odd numbers can be shown as set like this:
  - {..... -5, -3, -1, 1, 3, ...}
- Zero is considered an even number.

To tell whether a number is even or odd, look at the number in the ones place. That single number will tell you whether the entire number is odd or even.

- An even number ends in 0, 2, 4, 6 or 8
- An odd number ends in 1, 3, 5, 7 or 9

Example 4

1) Put a circle around the EVEN numbers:

1,289 3,206 647 1,648 2,611

2)Put a circle around the ODD numbers:

1,678 3,860 216 473 3,385 221

Basic Arithmetic Operations

The four basic arithmetic operators are add (+), subtract (−), multiply (×) and divide (÷)

Problem 1. Determine 735 + 167

H	T	U
	7	3 5
+	1	6 7
9	0	2
	1	1

(i) 5 + 7 = 12. Place 2 in units (U) column. Carry 1 in the tens (T) column.

(ii) 3 + 6 + 1 (carried) = 10. Place the 0 in the tens column. Carry the 1 in the hundreds (H) column.

(iii) 7 + 1 + 1 (carried) = 9. Place the 9 in the hundreds column.

Hence, **735 + 167 = 902**

Problem 2. Determine 632 − 369

H	T	U
	6	3 2
−	3	6 9
	2	6 3

(i) 2 − 9 is not possible; therefore 'borrow' 1 from the tens column (leaving 2 in the tens column). In the units column, this gives us 12 − 9 = 3.

(ii) Place 3 in the units column.

(iii) 2 − 6 is not possible; therefore 'borrow' 1 from the hundreds column (leaving 5 in the hundreds column). In the tens column, this gives us 12 − 6 = 6.

(iv) Place the 6 in the tens column.

Problem 7. Determine 86 × 7

H	T	U
	8	6
×		7
6	0	2
	4	

(i) 7 × 6 = 42. Place the 2 in the units (U) column and 'carry' the 4 into the tens (T) column.

(ii) 7 × 8 = 56; 56 + 4 (carried) = 60. Place the 0 in the tens column and the 6 in the hundreds (H) column.

Hence, 86 × 7 = 602

A good grasp of **multiplication tables** is needed when multiplying such numbers; a reminder of the multiplication table up to 12 × 12 is shown below. Confidence with handling numbers will be greatly improved if this table is memorized.

Problem 11. Determine 5796 ÷ 12

```
      483
   12)5796
      48
      99
      96
       36
       36
       00
```

(i) 12 into 5 won't go. 12 into 57 goes 4; place 4 above the 7 of 5796.

(ii) 4 × 12 = 48; place the 48 below the 57 of 5796

(iii) 57 − 48 = 9.

(iv) Bring down the 9 of 5796 to give 99.

(v) 12 into 99 goes 8; place 8 above the 9 of 5796.

(vi) 8 × 12 = 96; place 96 below the 99.

(vii) 99 − 96 = 3.

(viii) Bring down the 6 of 5796 to give 36.

(ix) 12 into 36 goes 3 exactly.

(x) Place the 3 above the final 6.

(xi) Place the 36 below the 36.

(xii) 36 − 36 = 0.

Hence, $5796 \div 12 = 5796/12 = \frac{5796}{12} = 483$.

The method shown is called **long division**.

Highest Common Factors

The **Highest Common Factor (HCF)** is the largest number which divides into two or more numbers exactly.

Example

12= 1, 2, 3, 4, 6 and 12 (i.e. all the numbers that divide into 12)

15= 1, 3, 5, and 15 (i.e. all the numbers that divide into 15)

- **1 and 3 are the only common factors; i.e., numbers which are factors of both 12 and 15**
- **Hence, the HCF of 12 and 15 is 3 since 3 is the highest number which divides into both 12 and 15**

Problem 12. Determine the HCF of the numbers 12, 30 and 42

Probably the simplest way of determining an HCF is to express each number in terms of its lowest factors. This is achieved by repeatedly dividing by the prime numbers 2, 3, 5, 7, 11, 13, … (where possible) in turn. Thus,

$$12 = 2 \times 2 \times 3$$
$$30 = 2 \times 3 \times 5$$
$$42 = 2 \times 3 \times 7$$

The factors which are common to each of the numbers are 2 in column 1 and 3 in column 3, shown by the broken lines. Hence, **the HCF is 2×3**; i.e., **6**. That is, 6 is the largest number which will divide into 12, 30 and 42.

Lowest Common Factors

A multiple is a number which contains another number an exact number of times. The **smallest number which is exactly divisible by each of two or more numbers is called the Lowest Common Multiple (LCM).**

Example

- The multiples of 12 are 12, 24, 36, 48, 60, 72,. . . and the multiples of 15 are 15, 30, 45, 60, 75,...
- 60 is a common multiple (i.e. a multiple of both 12 and 15) and there are no lower common multiples.

Hence, the LCM of 12 and 15 is 60 since 60 is the lowest number that both 12 and 15 divide into.

Problem 15. Determine the LCM of the numbers 150, 210, 735 and 1365

Using the method shown in Problem 14 above:

$150 = \boxed{2} \times \boxed{3} \times \boxed{5 \times 5}$

$210 = 2 \times 3 \times 5 \times 7$

$735 = 3 \times 5 \times \boxed{7 \times 7}$

$1365 = 3 \times 5 \times 7 \times \boxed{13}$

Hence, the LCM is $2 \times 3 \times 5 \times 5 \times 7 \times 7 \times 13 = 95550$.

Fraction

- A mathematical expression representing the division of one whole number by another where number usually expressed in the form a/b
    - Example : A mark of 9 out of 14 in an examination may be written as or 9/14.
- The number above the line, i.e. 9, is called the **numerator.** The number below the line, i.e. 14, is called the **denominator.**
- When the value of the numerator is less than the value of the denominator, the fraction is called a proper fraction. 9/14 is an example of a proper fraction.
- When the value of the numerator is greater than the value of the denominator, the fraction is called an improper fraction. 5/2 is an example of an **improper fraction.**
- A mixed number is a combination of a whole number and a fraction. is an example of a mixed number.

I. Adding & Subtracting Fractions

Before adding or subtracting fractions, we should make sure that the denominators are equal. Hence, the addition and subtraction of fractions are possible with a common denominator.

Example 6

Problem 1. Simplify $\frac{1}{3}+\frac{2}{7}$

$$\frac{1}{3}+\frac{2}{7}=\frac{1}{3}\times\frac{7}{7}+\frac{2}{7}\times\frac{3}{3}=\frac{7}{21}+\frac{6}{21}$$

$$=\frac{7+6}{21}=\mathbf{\frac{13}{21}}$$

Problem 2. Find the value of $3\frac{2}{3}-2\frac{1}{6}$

$$3\frac{2}{3}-2\frac{1}{6}=\left(3+\frac{2}{3}\right)-\left(2+\frac{1}{6}\right)$$

$$=3+\frac{2}{3}-2-\frac{1}{6}$$

$$=1+\frac{4}{6}-\frac{1}{6}=1\frac{3}{6}=\mathbf{1\frac{1}{2}}$$

Problem 3. Determine the value of

$$4\frac{5}{8}-3\frac{1}{4}+1\frac{2}{5}$$

$$4\frac{5}{8}-3\frac{1}{4}+1\frac{2}{5}=(4-3+1)+\left(\frac{5}{8}-\frac{1}{4}+\frac{2}{5}\right)$$

$$=2+\frac{5\times5-10\times1+8\times2}{40}$$

$$=2+\frac{25-10+16}{40}$$

$$=2+\frac{31}{40}=\mathbf{2\frac{31}{40}}$$

2. (a) $\frac{2}{7}+\frac{3}{11}$ (b) $\frac{2}{9}-\frac{1}{7}+\frac{2}{3}$

$$\left[\text{(a) } \frac{43}{77} \quad \text{(b) } \frac{47}{63}\right]$$

II. Multiplication & Division of Fraction

When we multiply two fractions, then the numerators are multiplied as well as the denominators are multiplied. Later simplify the fraction.

Example 7

Multiplication

To multiply two or more fractions together, the numerators are first multiplied to give a single number and this becomes the new numerator of the combined fraction. The denominators are then multiplied together to give the new denominator of the combined fraction.

For example, $\frac{2}{3} \times \frac{4}{7} = \frac{2 \times 4}{3 \times 7} = \frac{8}{21}$

Problem 9. Simplify $7 \times \frac{2}{5}$

$$7 \times \frac{2}{5} = \frac{7}{1} \times \frac{2}{5} = \frac{7 \times 2}{1 \times 5} = \frac{14}{5} = 2\frac{4}{5}$$

Division

The simple rule for division is **change the division sign into a multiplication sign and invert the second fraction.**

For example, $\frac{2}{3} \div \frac{3}{4} = \frac{2}{3} \times \frac{4}{3} = \frac{8}{9}$

Problem 14. Simplify $\frac{3}{7} \div \frac{8}{21}$

$$\frac{3}{7} \div \frac{8}{21} = \frac{3}{7} \times \frac{21}{8} = \frac{3}{1} \times \frac{3}{8} \text{ by cancelling}$$

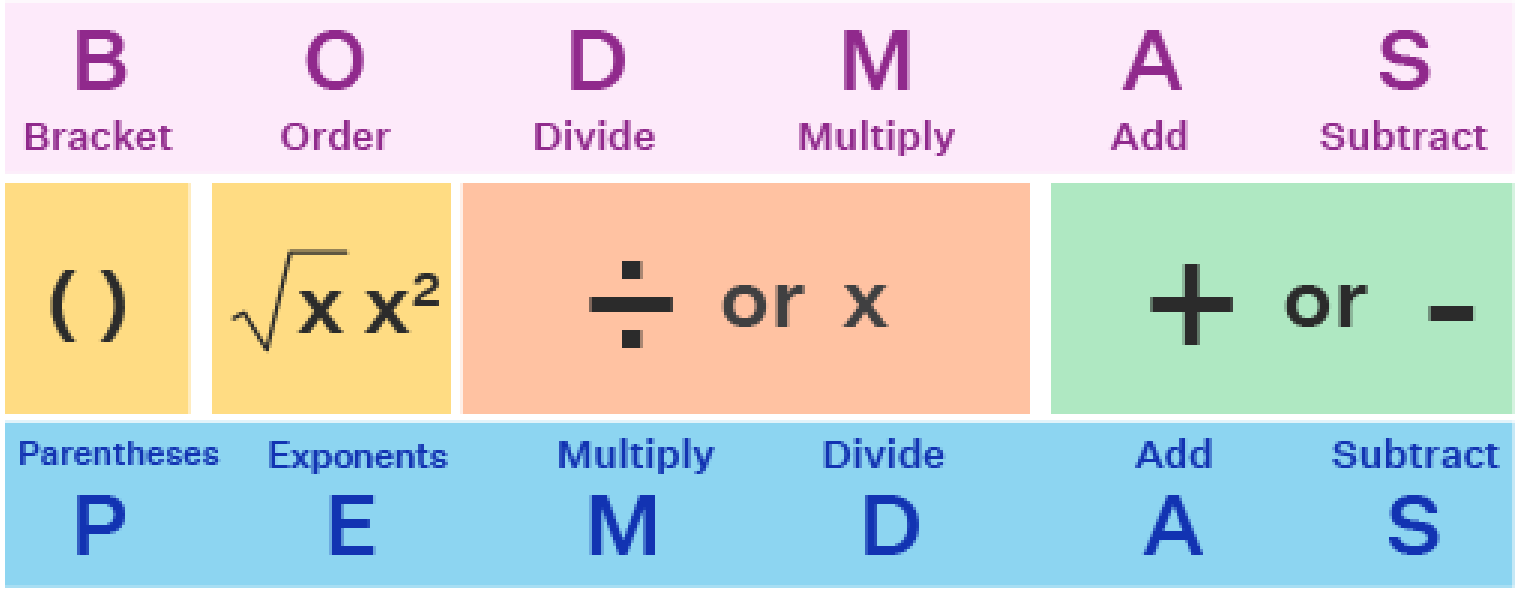
$$= \frac{3 \times 3}{1 \times 8} = \frac{9}{8} = 1\frac{1}{8}$$

Problem 15. Find the value of $5\frac{3}{5} \div 7\frac{1}{3}$

Operator Precedence

In mathematics, there is a definite order of precedence which we need to adhere to.

The order is as follows:



In practice, it does not matter if multiplication is performed before division or if subtraction is performed before addition., What is important is that the process of multiplication and division must be completed before addition and subtraction.

Example 8

$5 - 3 \times 4 + 24 \div (3 + 5) - 3^2$ is evaluated as follows:

$$5 - 3 \times 4 + 24 \div (3 + 5) - 3^2$$

$= 5 - 3 \times 4 + 24 \div 8 - 3^2$ (Bracket is removed and $3 + 5$ replaced with 8)

$= 5 - 3 \times 4 + 24 \div 8 - 9$ (Order means pOwer; in this case, $3^2 = 3 \times 3 = 9$)

$= 5 - 3 \times 4 + 3 - 9$ (Division: $24 \div 8 = 3$)

$= 5 - 12 + 3 - 9$ (Multiplication: $-3 \times 4 = -12$)

$= 8 - 12 - 9$ (Addition: $5 + 3 = 8$)

$= -13$ (Subtraction: $8 - 12 - 9 = -13$)

Precedence with the fraction

Problem 20. Evaluate

$$\frac{1}{3} \text{ of } \left(5\frac{1}{2} - 3\frac{3}{4}\right) + 3\frac{1}{5} \div \frac{4}{5} - \frac{1}{2}$$

$$\frac{1}{3} \text{ of } \left(5\frac{1}{2} - 3\frac{3}{4}\right) + 3\frac{1}{5} \div \frac{4}{5} - \frac{1}{2}$$

$= \frac{1}{3} \text{ of } 1\frac{3}{4} + 3\frac{1}{5} \div \frac{4}{5} - \frac{1}{2}$ (B)

$= \frac{1}{3} \times \frac{7}{4} + \frac{16}{5} \div \frac{4}{5} - \frac{1}{2}$ (O)

(Note that the 'of ' is replaced with a multiplication sign.)

$= \frac{1}{3} \times \frac{7}{4} + \frac{16}{5} \times \frac{5}{4} - \frac{1}{2}$ (D)

$= \frac{1}{3} \times \frac{7}{4} + \frac{4}{1} \times \frac{1}{1} - \frac{1}{2}$ by cancelling

$= \frac{7}{12} + \frac{4}{1} - \frac{1}{2}$ (M)

$= \frac{7}{12} + \frac{48}{12} - \frac{6}{12}$ (A/S)

$= \frac{49}{12}$

$= 4\frac{1}{12}$

Decimals

- Decimal comes from the Latin word **decimus**, meaning tenth, from the root word decem, or 10. The decimal system, therefore, has 10 as its base and is sometimes called a base-10 system.
- A decimal is a fraction written in a special form. Instead of writing 1/2, for example, you can express the fraction as the decimal 0.5, where the zero is in the one's place and the five is in the tenth place.
- **The decimal system of numbers is based on the digits 0 to 9.**
- There are a number of everyday occurrences in which we use decimal numbers.

Example

A radio is, say, tuned to 99.5MHz FM; 99.5 is an example of a decimal number

Converting Decimals to Fractions & Vice-versa

Problem 1. Convert 0.375 to a proper fraction in its simplest form

(i) 0.375 may be written as $\frac{0.375 \times 1000}{1000}$ i.e. $0.375 = \frac{375}{1000}$

(ii) Dividing both numerator and denominator by 5 gives $\frac{375}{1000} = \frac{75}{200}$

(iii) Dividing both numerator and denominator by 5 again gives $\frac{75}{200} = \frac{15}{40}$

(iv) Dividing both numerator and denominator by 5 again gives $\frac{15}{40} = \frac{3}{8}$

Since both 3 and 8 are only divisible by 1, we cannot 'cancel' any further, so $\frac{3}{8}$ is the 'simplest form' of the fraction.

Hence, the decimal fraction $0.375 = \frac{3}{8}$ as a proper fraction.

Problem 3. Express $\frac{7}{8}$ as a decimal fraction

To convert a proper fraction to a decimal fraction, the numerator is divided by the denominator.

$$8\overline{)7.000}\quad 0.875$$

(i) 8 into 7 will not go. Place the 0 above the 7.

(ii) Place the decimal point above the decimal point of 7.000

(iii) 8 into 70 goes 8, remainder 6. Place the 8 above the first zero after the decimal point and carry the 6 remainder to the next digit on the right, making it 60.

(iv) 8 into 60 goes 7, remainder 4. Place the 7 above the next zero and carry the 4 remainder to the next digit on the right, making it 40.

(v) 8 into 40 goes 5, remainder 0. Place 5 above the next zero.

Hence, the proper fraction $\frac{7}{8} = 0.875$ as a decimal fraction.

Problem 2. Convert 3.4375 to a mixed number

(i) 0.4375 may be written as $\frac{0.4375 \times 10000}{10000}$ i.e. $0.4375 = \frac{4375}{10000}$

(ii) Dividing both numerator and denominator by 25 gives $\frac{4375}{10000} = \frac{175}{400}$

(iii) Dividing both numerator and denominator by 5 gives $\frac{175}{400} = \frac{35}{80}$

(iv) Dividing both numerator and denominator by 5 again gives $\frac{35}{80} = \frac{7}{16}$

Since both 5 and 16 are only divisible by 1, we cannot 'cancel' any further, so $\frac{7}{16}$ is the 'lowest form' of the fraction.

(v) Hence, $0.4375 = \frac{7}{16}$

Thus, the decimal fraction $3.4375 = 3\frac{7}{16}$ as a mixed number.

Significant Figures and Decimal Places

- **Terminating decimal**

A number that can be expressed exactly as a decimal fraction.

$$3\frac{3}{16} = 3.1825 \text{ is a terminating decimal}$$

- **Non-terminating decimal**

A number that cannot be expressed exactly as a decimal fraction.

$$1\frac{5}{7} = 1.7142857\ldots \text{ is a non-terminating decimal}$$

The answer to a non-terminating decimal may be expressed in two ways, depending on the accuracy required:

(a) correct to a number of significant figures, or

$$1.7142857\ldots = \mathbf{1.7143} \text{ correct to 5 significant figures}$$

(b) correct to a number of decimal places i.e. the number of figures after the decimal point

$$1.7142857\ldots = \mathbf{1.7143} \text{ correct to 4 decimal places}$$

Adding and Subtracting Decimal Numbers

Problem 7. Evaluate $46.8 + 3.06 + 2.4 + 0.09$ and give the answer correct to 3 significant figures

The decimal points are placed under each other as shown. Each column is added, starting from the right.

$$\begin{array}{r} 46.8 \\ 3.06 \\ 2.4 \\ +0.09 \\ \hline 52.35 \\ 1\,1\;1 \end{array}$$

(i) $6 + 9 = 15$. Place 5 in the hundredths column. Carry 1 in the tenths column.

(ii) $8 + 0 + 4 + 0 + 1$ (carried) $= 13$. Place the 3 in the tenths column. Carry the 1 into the units column.

(iii) $6 + 3 + 2 + 0 + 1$ (carried) $= 12$. Place the 2 in the units column. Carry the 1 into the tens column.

(iv) $4 + 1$(carried) $= 5$. Place the 5 in the hundreds column.

Hence,

$$46.8 + 3.06 + 2.4 + 0.09 = 52.35$$
$$= 52.4, \text{ correct to 3 significant figures}$$

Problem 8. Evaluate $64.46 - 28.77$ and give the answer correct to 1 decimal place

As with addition, the decimal points are placed under each other as shown.

$$\begin{array}{r} 64.46 \\ -28.77 \\ \hline 35.69 \end{array}$$

(i) $6 - 7$ is not possible; therefore 'borrow' 1 from the tenths column. This gives $16 - 7 = 9$. Place the 9 in the hundredths column.

(ii) $3 - 7$ is not possible; therefore 'borrow' 1 from the units column. This gives $13 - 7 = 6$. Place the 6 in the tenths column.

(iii) $3 - 8$ is not possible; therefore 'borrow' from the hundreds column. This gives $13 - 8 = 5$. Place the 5 in the units column.

(iv) $5 - 2 = 3$. Place the 3 in the hundreds column.

Hence,

$$64.46 - 28.77 = 35.69$$
$$= 35.7 \text{ correct to 1 decimal place}$$

Multiplying and Dividing Decimal Numbers

Problem 11. Evaluate $44.25 \div 1.2$, correct to (a) 3 significant figures, (b) 2 decimal places

$$44.25 \div 1.2 = \frac{44.25}{1.2}$$

The denominator is multiplied by 10 to change it into an integer. The numerator is also multiplied by 10 to keep the fraction the same. Thus,

$$\frac{44.25}{1.2} = \frac{44.25 \times 10}{1.2 \times 10} = \frac{442.5}{12}$$

$$\begin{array}{r} 36.875 \\ 12\overline{)442.500} \\ \underline{36} \\ 82 \\ \underline{72} \\ 105 \\ \underline{96} \\ 90 \\ \underline{84} \\ 60 \\ \underline{60} \\ 0 \end{array}$$

(i) 12 into 44 goes 3; place the 3 above the second 4 of 442.500

(ii) $3 \times 12 = 36$; place the 36 below the 44 of 442.500

iii) $44 - 36 = 8$.

iv) Bring down the 2 to give 82.

(v) 12 into 82 goes 6; place the 6 above the 2 of

(vi) $6 \times 12 = 72$; place the 72 below the 82.

(vii) $82 - 72 = 10$.

(viii) Bring down the 5 to give 105.

(ix) 12 into 105 goes 8; place the 8 above the 5 of 442.500

(x) $8 \times 12 = 96$; place the 96 below the 105.

(xi) $105 - 96 = 9$.

(xii) Bring down the 0 to give 90.

(xiii) 12 into 90 goes 7; place the 7 above the first zero of 442.500

(xiv) $7 \times 12 = 84$; place the 84 below the 90.

(xv) $90 - 84 = 6$.

(xvi) Bring down the 0 to give 60.

(xvii) 12 into 60 gives 5 exactly; place the 5 above the second zero of 442.500

(xviii) Hence, $44.25 \div 1.2 = \frac{442.5}{12} = \mathbf{36.875}$

Problem 10. Evaluate 37.6×5.4

$$\begin{array}{r} 376 \\ \times 54 \\ \hline 1504 \\ \underline{18800} \\ 20304 \end{array}$$

(i) $376 \times 54 = 20304$.

(ii) As there are $1 + 1 = 2$ digits to the right of the decimal points of the two numbers being multiplied together, $37.\underline{6} \times 5.\underline{4}$, then

$$\mathbf{37.6 \times 5.4 = 203.04}$$

Percentage

A **percentage** is a number or ratio expressed as a fraction of 100. It is often denoted using the **percent** sign, "%", or the abbreviation "pct".

Percentages are fractions with 100 as their denominator.

Example

Fraction 40/100 = 40% (forty percent)

❖ Convert a decimal to a percentage & convert a percentage to a decimal

A decimal number is converted to a percentage by multiplying by 100.

Problem 1. Express 0.015 as a percentage

To express a decimal number as a percentage, merely multiply by 100, i.e.

$$0.015 = 0.015 \times 100\%$$
$$= \mathbf{1.5\%}$$

Multiplying a decimal number by 100 means moving the decimal point 2 places **to the right**.

A percentage is converted to a decimal number by dividing by 100.

Problem 3. Express 6.5% as a decimal number

$$6.5\% = \frac{6.5}{100} = \mathbf{0.065}$$

Dividing by 100 means moving the decimal point 2 places **to the left**.

❖ Convert a fraction to a percentage

A fraction is converted to a percentage by multiplying by 100.

Problem 5. Express $\frac{5}{8}$ as a percentage

$$\frac{5}{8} = \frac{5}{8} \times 100\% = \frac{500}{8}\%$$
$$= 62.5\%$$

❖ Expressing one quantity as a percentage of another quantity

To express one quantity as a percentage of another quantity, divide the first quantity by the second then multiply by 100.

Problem 13. Express 23 cm as a percentage of 72 cm, correct to the nearest 1%

$$23\,\text{cm as a percentage of } 72\,\text{cm} = \frac{23}{72} \times 100\%$$
$$= 31.94444\ldots\%$$
$$= 32\% \text{ correct to the nearest } 1\%$$

❖ Convert a percentage to a fraction

A percentage is converted to a fraction by dividing by 100 and then, by cancelling, reducing it to its simplest form.

Problem 8. Express 75% as a fraction

$$75\% = \frac{75}{100}$$
$$= \frac{3}{4}$$

The fraction $\frac{75}{100}$ is reduced to its simplest form by cancelling, i.e. dividing both numerator and denominator by 25.

❖ Finding a percentage of a quantity

To find a percentage of a quantity, convert the percentage to a fraction (by dividing by 100) and remember that 'of' means multiply.

Problem 10. Find 27% of £65

$$27\% \text{ of } £65 = \frac{27}{100} \times 65$$
$$= £17.55 \text{ by calculator}$$

❖ Percentage change

Percentage change is given by

$$\frac{\text{new value} - \text{original value}}{\text{original value}} \times 100\%.$$

Problem 15. A box of resistors increases in price from £45 to £52. Calculate the percentage change in cost, correct to 3 significant figures

$$\%\ \text{change} = \frac{\text{new value} - \text{original value}}{\text{original value}} \times 100\%$$

$$= \frac{52-45}{45} \times 100\% = \frac{7}{45} \times 100$$

$$= \mathbf{15.6\%} = \textbf{percentage change in cost}$$

❖ Percentage error

Problem 18. The voltage across a component in an electrical circuit is calculated as 50 V using Ohm's law. When measured, the actual voltage is 50.4 V. Calculate, correct to 2 decimal places, the percentage error in the calculation

$$\%\ \text{error} = \frac{\text{error}}{\text{correct value}} \times 100\%$$

$$= \frac{50.4-50}{50.4} \times 100\%$$

$$= \frac{0.4}{50.4} \times 100\% = \frac{40}{50.4}\%$$

$$= \mathbf{0.79\%}$$

The percentage error in the calculation is **0.79% too low**, which is sometimes written as **−0.79% error**.

❖ Original Value

$$\text{Original value} = \frac{\text{new value}}{100 \pm \%\ \text{change}} \times 100\%$$

Problem 19. A man pays £149.50 in a sale for a DVD player which is labelled '35% off'. What was the original price of the DVD player?

In this case, it is a 35% reduction in price, so we use $\frac{\text{new value}}{100 - \%\ \text{change}} \times 100$, i.e. a minus sign in the denominator.

$$\text{Original price} = \frac{\text{new value}}{100 - \%\ \text{change}} \times 100$$

$$= \frac{149.5}{100-35} \times 100$$

$$= \frac{149.5}{65} \times 100 = \frac{14950}{65}$$

$$= \mathbf{£230}$$

❖ Percentage increase/ decrease and interest

$$\text{New value} = \frac{100 + \%\ \text{increase}}{100} \times \text{original value}$$

Problem 22. £3600 is placed in an ISA account which pays 6.25% interest per annum. How much is the investment worth after 1 year?

$$\text{Value after 1 year} = \frac{100+6.25}{100} \times £3600$$

$$= \frac{106.25}{100} \times £3600$$

$$= 1.0625 \times £3600$$

$$= \mathbf{£3825}$$

❑ Commutative Properties (Addition/Multiplication)

❖ When JUST adding or multiplying (not at the same time), you can move the numbers around and the answer will be the same.

Addition

$$A + B + C = B + A + C$$
$$7 + 17 + 23 = 23 + 7 + 17$$

Multiplication

$$A \cdot B \cdot C = B \cdot A \cdot C$$
$$11 \cdot 5 \cdot 3 = 5 \cdot 3 \cdot 11$$

❑ Associative Properties (Addition/Multiplication)

❖ When JUST adding or multiplying (not at the same time), you can change the grouping symbol locations and get the same answer.

Addition

$$A + (B + C) = (A + B) + C$$
$$(3 + 4) + 7 = 3 + (4 + 7)$$

Multiplication

$$A + (B + C) = (A + B) + C$$
$$(3 + 4) + 7 = 3 + (4 + 7)$$

❑ Identity Properties of Addition and Multiplication

- The identity for ADDITION is ZERO.
- Addition likes ZERO.

Example $5 + 0 = 5$

The identity for MULTIPLICATION is ONE

Multiplication likes ONE.

Example $4 \cdot 1 = 4$

Identify the properties shown	
3 + 6= 6 + 3	Commutative Property
8 = 1 • 8	Multiplicative Identity
(3z)m = 3(zm)	Associative Property of Multiplication
0 + b = b	Additive Identity

Summary of Properties of Real Numbers

The real number system is an example of a mathematical structure called a field Some of the properties of a field are summarized in the table below:

Real Number Properties		
For any real numbers **a**, **b**, and **c**.		
Property	Addition	Multiplication
Commutative	$a+b=b+a$	$a\cdot b=b\cdot a$
Associative	$(a+b)+c=a+(b+c)$	$(a\cdot b)\cdot c=a\cdot(b\cdot c)$
Identity	$a+0=a=0+a$	$a\cdot 1=a=1\cdot a$
Inverse	$a+(-a)=0=(-a)+a$	If $a\neq 0$, then $a\cdot\frac{1}{a}=1=\frac{1}{a}\cdot a$
Distributive	$a(b+c)=ab+ac$ and	$(b+c)a=ba+ca$

Thank you